ESSENTIAL SURJECTIVITY AND FULL FAITHFULNESS TRANSFER BETWEEN SIBLINGS

THE STACKS PROJECT AUTHORS

Editorial note. One named auxiliary 'details omitted' (five-lemma Hom check) transcribed as-is; the main Ext induction is complete in-span.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2291 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Sibling definition and property (2). definition-siblings (same chapter, not carded). Gives the functorial isomorphisms a_X and the factorization used throughout.
- Five-lemma and representable-homological. homology-lemma-five-lemma, derived-lemma-representable-homological (cross-chapter, named). Feeds the auxiliary Hom-isomorphism extension step.
- Split and Yoneda extension machinery. derived-lemma-split, derived-section-ext (cross-chapter, named). Split detection and the q>1 Yoneda composition decomposition.

Source reference. Original definition 0FZT: sources/stacks-equiv/pinned-original-source.tex, source lines 2189–2189. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0149: focused-seeds/geometry/S9-eqv-sibling-faithful/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QT: focused-seeds/geometry/S9-eqv-sibling-faithful/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06XP: focused-seeds/geometry/S9-eqv-sibling-faithful/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05QB: focused-seeds/geometry/S9-eqv-sibling-faithful/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let F and F' be siblings as in Definition 0FZT. Then*

(1) *if F is essentially surjective, then F' is essentially surjective,*
(2) *if F is fully faithful, then F' is fully faithful.*

Proof. Part (1) is immediate from property (2) for siblings.

Assume F is fully faithful. Denote $\mathcal{D}' \subset \mathcal{D}$ the essential image of F so that $F : D^b(\mathcal{A}) \to \mathcal{D}'$ is an equivalence. Since the functor F' factors through $\mathcal{D}'$ by property (2) for siblings, we can consider the functor $H = F^{-1} \circ F' : D^b(\mathcal{A}) \to D^b(\mathcal{A})$. Observe that H is a sibling of the identity functor. Since it suffices to prove that H is fully faithful, we reduce to the problem discussed in the next paragraph.

Set $\mathcal{D} = D^b(\mathcal{A})$. We have to show a sibling $F : \mathcal{D} \to \mathcal{D}$ of the identity functor is fully faithful. Denote $a_X : X \to F(X)$ the functorial isomorphism for $X \in \mathrm{Ob}(\mathcal{A})$ given to us by Definition 0FZT. For any K in $\mathcal{D}$ and distinguished triangle $K_1 \to K_2 \to K_3$ of $\mathcal{D}$ if the maps

$$F : \mathrm{Hom}(K, K_i[n]) \to \mathrm{Hom}(F(K), F(K_i[n]))$$

are isomorphisms for all $n \in \mathbf{Z}$ and $i = 1, 3$, then the same is true for $i = 2$ and all $n \in \mathbf{Z}$. This uses the 5-lemma Homology, Lemma 05QB and Derived Categories, Lemma 0149; details omitted. Similarly, if the maps

$$F : \mathrm{Hom}(K_i[n], K) \to \mathrm{Hom}(F(K_i[n]), F(K))$$

are isomorphisms for all $n \in \mathbf{Z}$ and $i = 1, 3$, then the same is true for $i = 2$ and all $n \in \mathbf{Z}$. Using the canonical truncations and induction on the number of nonzero cohomology objects, we see that it is enough to show

$$F : \mathrm{Ext}^q(X, Y) \to \mathrm{Ext}^q(F(X), F(Y))$$

is bijective for all $X, Y \in \mathrm{Ob}(\mathcal{A})$ and all $q \in \mathbf{Z}$. Since F is a sibling of id we have $F(X) \cong X$ and $F(Y) \cong Y$ hence the right hand side is zero for $q < 0$. The case $q = 0$ is OK by our assumption that F is a sibling of the identity functor. It remains to prove the cases $q > 0$.

The case $q = 1$: Injectivity. An element ξ of $\mathrm{Ext}^1(X, Y)$ gives rise to a distinguished triangle

$$Y \to E \to X \xrightarrow{\xi} Y[1]$$

Observe that $E \in \mathrm{Ob}(\mathcal{A})$. Since F is a sibling of the identity functor we obtain a commutative diagram

$$\begin{array}{ccc} E & \longrightarrow & X \\ \downarrow & & \downarrow \\ F(E) & \longrightarrow & F(X) \end{array}$$

whose vertical arrows are the isomorphisms a_E and a_X. By TR3 the distinguished triangle associated to ξ we started with is isomorphic to the distinguished triangle

$$F(Y) \to F(E) \to F(X) \xrightarrow{F(\xi)} F(Y[1]) = F(Y)[1]$$

Thus $\xi = 0$ if and only if $F(\xi)$ is zero, i.e., we see that $F : \mathrm{Ext}^1(X, Y) \to \mathrm{Ext}^1(F(X), F(Y))$ is injective.

The case $q = 1$: Surjectivity. Let θ be an element of $\mathrm{Ext}^1(F(X), F(Y))$. This defines an extension of $F(X)$ by $F(Y)$ in $\mathcal{A}$ which we may write as $F(E)$ as F is a sibling of the identity functor. We thus get a distinguished triangle

$$F(Y) \xrightarrow{F(\alpha)} F(E) \xrightarrow{F(\beta)} F(X) \xrightarrow{\theta} F(Y[1]) = F(Y)[1]$$

for some morphisms $\alpha : Y \to E$ and $\beta : E \to X$. Since F is a sibling of the identity functor, the sequence $0 \to Y \to E \to X \to 0$ is a short exact sequence in $\mathcal{A}$! Hence we obtain a distinguished triangle

$$Y \xrightarrow{\alpha} E \xrightarrow{\beta} X \xrightarrow{\delta} Y[1]$$

for some morphism $\delta : X \to Y[1]$. Applying the exact functor F we obtain the distinguished triangle

$$F(Y) \xrightarrow{F(\alpha)} F(E) \xrightarrow{F(\beta)} F(X) \xrightarrow{F(\delta)} F(Y)[1]$$

Arguing as above, we see that these triangles are isomorphic. Hence there exists a commutative diagram

$$\begin{array}{ccc} F(X) & \xrightarrow[F(\delta)]{} & F(Y[1]) \\ \downarrow{\scriptstyle \gamma} & & {\scriptstyle \epsilon}\downarrow \\ F(X) & \xrightarrow{\theta} & F(Y[1]) \end{array}$$

for some isomorphisms γ, ϵ (we can say more but we won't need more information). We may write $\gamma = F(\gamma')$ and $\epsilon = F(\epsilon')$. Then we have $\theta = F(\epsilon' \circ \delta \circ (\gamma')^{-1})$ and we see the surjectivity holds.

The case $q > 1$: surjectivity. Using Yoneda extensions, see Derived Categories, Section 06XP, we find that for any element ξ in $\mathrm{Ext}^q(F(X), F(Y))$ we can find $F(X) = B_0, B_1, \ldots, B_{q-1}, B_q = F(Y) \in \mathrm{Ob}(\mathcal{A})$ and elements

$$\xi_i \in \mathrm{Ext}^1(B_{i-1}, B_i)$$

such that ξ is the composition $\xi_q \circ \ldots \circ \xi_1$. Write $B_i = F(A_i)$ (of course we have $A_i = B_i$ but we don't need to use this) so that

$$\xi_i = F(\eta_i) \in \mathrm{Ext}^1(F(A_{i-1}), F(A_i)) \quad \text{with} \quad \eta_i \in \mathrm{Ext}^1(A_{i-1}, A_i)$$

by surjectivity for $q = 1$. Then $\eta = \eta_q \circ \ldots \circ \eta_1$ is an element of $\mathrm{Ext}^q(X, Y)$ with $F(\eta) = \xi$.

The case $q > 1$: injectivity. An element ξ of $\mathrm{Ext}^q(X, Y)$ gives rise to a distinguished triangle

$$Y[q-1] \to E \to X \xrightarrow{\xi} Y[q]$$

Applying F we obtain a distinguished triangle

$$F(Y)[q-1] \to F(E) \to F(X) \xrightarrow{F(\xi)} F(Y)[q]$$

If $F(\xi) = 0$, then $F(E) \cong F(Y)[q-1] \oplus F(X)$ in $\mathcal{D}$, see Derived Categories, Lemma 05QT. Since F is a sibling of the identity functor we have $E \cong F(E)$ and hence

$$E \cong F(E) \cong F(Y)[q-1] \oplus F(X) \cong Y[q-1] \oplus X$$

In other words, E is isomorphic to the direct sum of its cohomology objects. This implies that the initial distinguished triangle is split, i.e., $\xi = 0$. $\square$